

## MODULE 2 : Differential Equations

+ what is | a differential equation?  
| an ordinary differential equation?

\* Examples ?

+ Solutions to differential equations  
Initial Value Problems (IVPs)

Numerical \*  
Beef!

BAŞAK ← This is me

Algebraic equations

$$f(y) = 0$$

$$\sin y + e^y + \pi = 0$$

$$ay + b = 0$$

Differential equations

$$\frac{dy(t)}{dt} = f(y(t), t)$$

$$\dot{y} = \frac{dy}{dt}$$

notation

Examples:

$$\dot{y} = ky \quad \text{or} \quad \dot{y} = -ky \quad \left( \begin{array}{l} \text{exponential} \\ \text{growth/decay} \end{array} \right)$$

$\nwarrow y = e^{-kt}$

$$\dot{y} = ky(y_{\max} - y)$$

sys. of  
diff. eqs

$$\dot{x} = ax - bxy$$

$$\dot{y} = -cy + dxy$$

} predator prey.

Consider  $\dot{y} = -2y$

Is  $y(t) = e^{-2t}$  a solution to this diff. eq<sup>n</sup>?

Let's try

$$\frac{dy(t)}{dt} = -2 \underbrace{e^{-2t}}_{y(t)} = -2y(t) \checkmark \text{ It is!}$$

Is  $y(t) = e^{-2t} + c$  a solution?

$$\frac{dy(t)}{dt} = -2 \underbrace{e^{-2t}}_{y(t)-c} \quad \times \text{ NO it's not}$$

Is  $y(t) = C e^{-2t}$

$$\frac{dy(t)}{dt} = -2 \underbrace{C e^{-2t}}_{y(t)} = -2y(t) \checkmark$$

A family of solutions

For the diff eq<sup>n</sup>  $\dot{y} = -2y$ ,  $y(t) = C e^{-2t}$  represents a class of solutions.

What if I know  $y(t_0) = y_0$ ?  
initial value

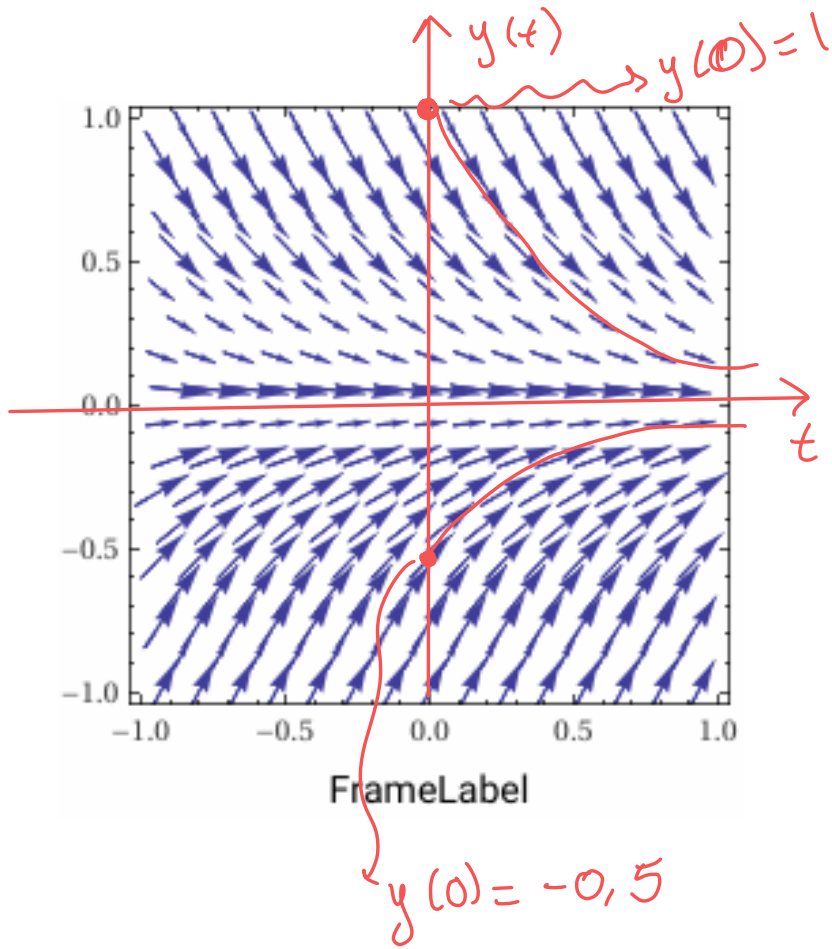
Then  $y(t_0) = \underline{C} e^{-2t_0} = y_0$   $\leadsto$  If  $t_0 = 0$   
 $\hookrightarrow$  can solve for this  $y(t) = y_0 e^{-2t}$

Initial value Problem (IVP):

Given  $\dot{y} = f(y, t)$  <sup>(a)</sup> and an initial condition, that is,  $y(t_0) = y_0$  <sup>(b)</sup>,

Find  $y: [t_0, t_f] \rightarrow \mathbb{R}$  such that  $y(t)$  satisfies (a) and (b)

for all  $t \in [t_0, t_f]$

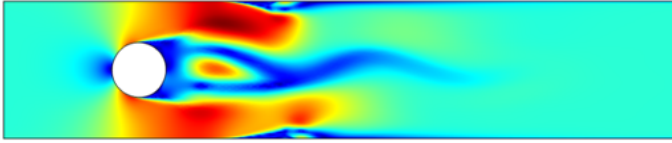
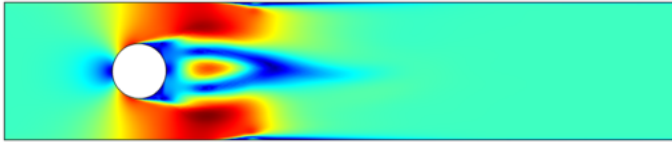


$$e^{-2t}$$

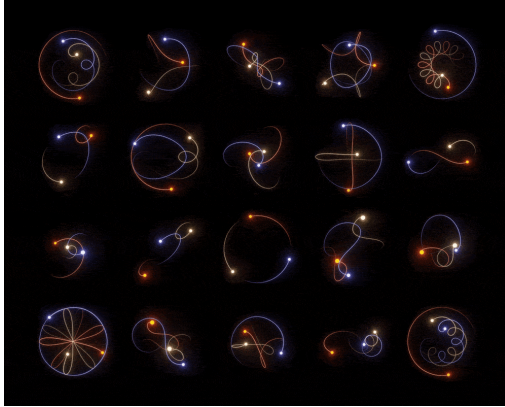
All the curves for  
 $C e^{-2t}$  for various  $C$

$$C e^{-2t_0} = y(t_0)$$

Examples → problems needing numerical solutions



← Flow of a fluid  
Navier-stokes



3 body  
problem

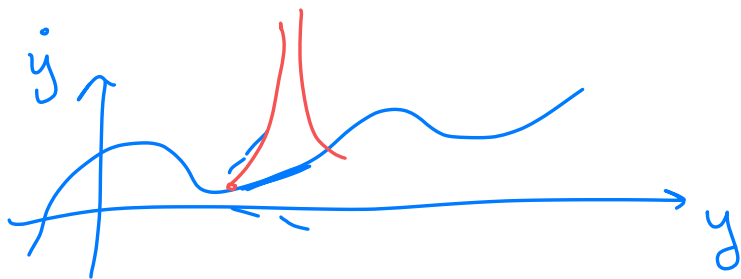
$$\begin{aligned}\ddot{\mathbf{r}}_1 &= -Gm_2 \frac{(\mathbf{r}_1 - \mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|^3} - Gm_3 \frac{(\mathbf{r}_1 - \mathbf{r}_3)}{|\mathbf{r}_1 - \mathbf{r}_3|^3}, \\ \ddot{\mathbf{r}}_2 &= -Gm_3 \frac{(\mathbf{r}_2 - \mathbf{r}_3)}{|\mathbf{r}_2 - \mathbf{r}_3|^3} - Gm_1 \frac{(\mathbf{r}_2 - \mathbf{r}_1)}{|\mathbf{r}_2 - \mathbf{r}_1|^3}, \\ \ddot{\mathbf{r}}_3 &= -Gm_1 \frac{(\mathbf{r}_3 - \mathbf{r}_1)}{|\mathbf{r}_3 - \mathbf{r}_1|^3} - Gm_2 \frac{(\mathbf{r}_3 - \mathbf{r}_2)}{|\mathbf{r}_3 - \mathbf{r}_2|^3}.\end{aligned}$$



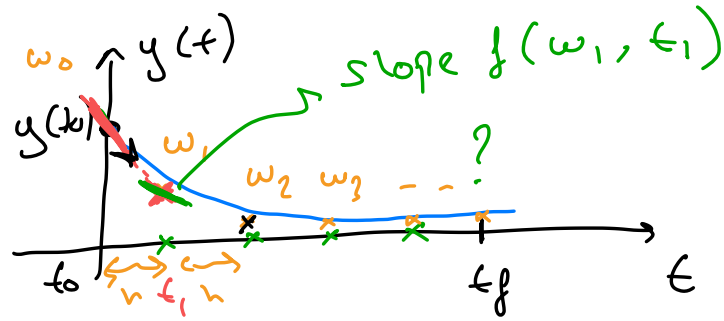
↑ ISAAC SIM  
NVIDIA

## Existence & Uniqueness

Given an IVP  $\dot{y} = f(y, t)$  and  $y(t_0) = y_0$ , suppose the partial derivative  $\frac{\partial f}{\partial y}$  exists and  $\left| \frac{\partial f}{\partial y} \right|$  is bounded, that is,  $\left| \frac{\partial f}{\partial y} \right| \leq L$  for all  $t_0 \leq t \leq t_f$  and all  $y$ . Then, IVP has a unique sol<sup>n</sup> for  $t \in [t_0, t_f]$ .



# Euler's Method



$$h = \frac{t_f - t_0}{N} \rightarrow \text{number of steps}$$

step size

$$w_{i+1} = w_i + h f(w_i, t_i)$$

$$t_i = t_{i-1} + h$$

start from  $w_0$

$$\left. \begin{aligned} \dot{y} &= f(y, t) \\ y(t_0) &= y_0 \\ y(t_f) &= ? \end{aligned} \right\} \text{known}$$

$$w_0 = y_0$$

$$w_1 = w_0 + h f(w_0, t_0)$$

approximates  $y(t_1)$

$$w_2 = w_1 + \underbrace{h f(w_1, t_1)}_{\text{slope}}$$

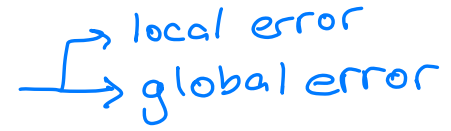
$\vdots$

$$w_n = w_{n-1} + h f(\dots)$$



How good is an approximation method?

We consider two types of errors

① what happens when  $h \rightarrow 0$ ? 

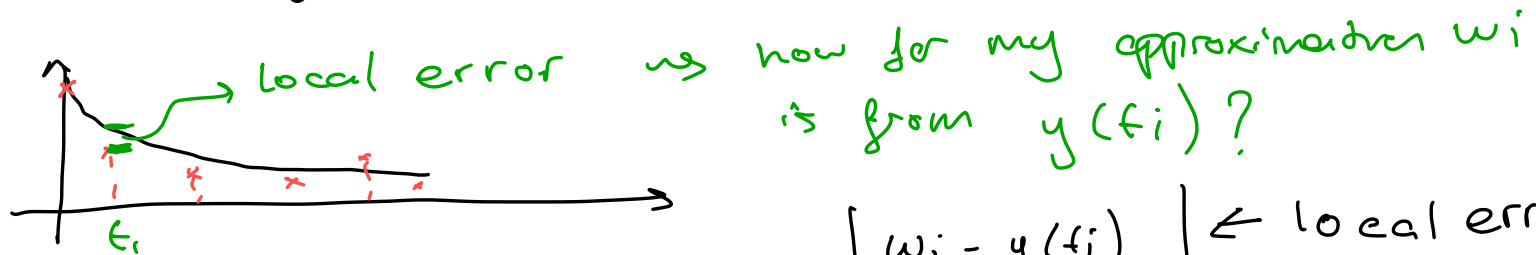
② what happens when  $h$  is fixed?

 does the solution remains in the neighborhood for a given  $h$

① is also called zero-stability

② is called absolute stability.

## Local vs global error

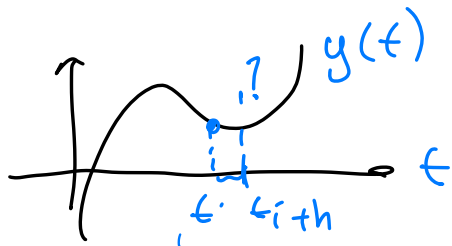


$$|w_i - y(t_i)| \leftarrow \text{local error}$$

Since  $w_i$  isn't exact  $w_{i+1}$  includes errors from previous steps.

Global error is the total error that accumulates for the solution and we can see it as

$$|y(t_f) - w_N|$$



$$y(t_i + h) = ?$$

✓ Taylor expansion  
(Module 3)

$$y(t_i + h) \stackrel{*}{=} \underbrace{y(t_i) + \dot{y}(t_i)h}_{\text{looks like Euler's method}} + \underbrace{\frac{1}{2} \ddot{y}(t_i)h^2 + \frac{1}{6} \dddot{y}(t_i)h^3 + \dots}_{\text{error}}$$

→ This is an equality not an approximation

Euler's method

$$y(t_i + h) \approx y(t_i) + h \underbrace{\dot{y}(t_i)}_{f(y(t_i), t_i)} = w_i$$

$$y(t_i + h) - w_i = \text{error}$$

$$\underline{\underline{O(h^2)}}$$

Local error  
for Euler's  
method

Big-O notation  $f(x) = O(g(x))$  error is  $O(h^2)$

$$f(x) \leq M g(x)$$

$$\text{error}(h) \leq M h^2$$

Check Thm 6.2.1 FNC book

Global error  $\rightarrow$  local error adds up @ each step

$$\text{number of steps } O(N) \quad O(h^2) = O(Nh^2)$$

$$N = \frac{t_f - t_0}{h} \quad O(1/h) \quad O(h^2) = \boxed{O(h)}$$

$\uparrow$  Global error of Euler's method

Order of a method:

If global error is  $O(h^p)$  then the method is called  $p^{\text{th}}$  order.

Euler's method is first order!

Example:  $\dot{y} = f(y, t) = e^{-t} - y^2$  w/  $y(0) = 0$   
 we want  $y(1)$ . use  $h = 0.5$   $y(t_0) = y_0$   
 $\downarrow$   $\downarrow$   
 $0$   $0$

$w_{i+1} = w_i + h f(w_i, t_i) \rightarrow$  Euler's method

$$w_0 = y(0) = 0$$

$$w_1 = w_0 + 0.5 (e^{-0} - 0) = 0.5 \quad t_1 = t_0 + h = h = 0.5$$

$$w_2 = w_1 + 0.5 (e^{-0.5} - 0.5^2) = 0.57 \dots = \boxed{0.678} \quad t_2 = t_1 + h = 1$$

$$y(1) \approx w_2$$

$$y(1) = 0.503347 \quad \text{Exact value}$$

absolute error  $\approx 0.17$   $\rightarrow$  try with  $h = 1$   
 $h = 0.2 \rightarrow$  does the error match your estimate?

Expectations.

$$h = 0.5 \quad \text{error} \approx 0.17$$

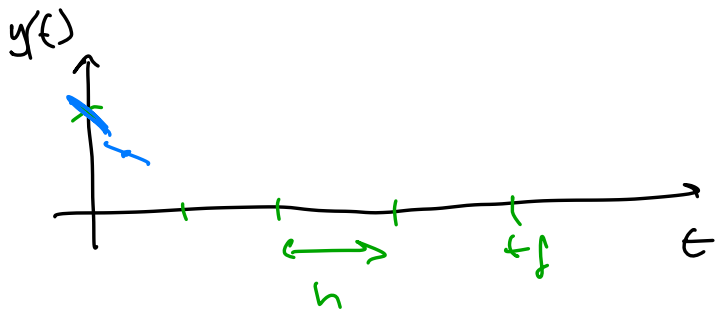
$$h = 0.25 \quad \text{error ?} \quad \leadsto \approx \frac{0.17}{2}$$

Global error  $O(h)$

Euler's method

Can we do better?

↳ improve our approximation




 } linear combination  
of intermediary  
slopes

Consider the family

$$y(t+h) \approx y(t) + h(a_1 k_1 + a_2 k_2)$$

$$k_1 = f(y(t), t)$$

$$k_2 = f(y(t) + \alpha k_1 h, t + \beta h)$$

If  $a_1 = 1$  &  $a_2 = 0 \rightarrow$  Euler's method

But I want something at least  $O(h^2)$

How can I pick  $a_1, a_2, \alpha, \beta$  so the global error is  $O(h^2)$ ?

Remember

$$y(t_i+h) \stackrel{*}{=} y(t_i) + \dot{y}(t_i)h + \frac{1}{2}\ddot{y}(t_i)h^2 + \frac{1}{6}\ddot{\ddot{y}}(t_i)h^3 + \dots$$

Euler's  
method was  
accurate until  
here

We need to be accurate  
until here for global  
error to be  $O(h^2)$ .

$O(h^3) \rightarrow$  local error  
 $O(h^2) \rightarrow$  global error.

Solving this for matching until this point gives  
(not examinable)

$\alpha \rightarrow$  free variable  $\alpha \neq 0$

$$\beta = \alpha$$

$$a_1 = 1 - \frac{1}{2\alpha}$$

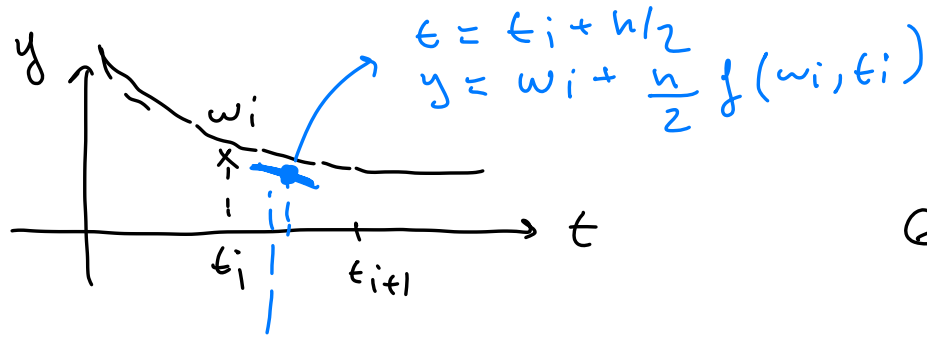
$$a_2 = \frac{1}{2\alpha}$$



## Midpoint Method

Pick  $\alpha = 1/2 \rightarrow \alpha_1 = 0 \quad \alpha_2 = 1 \quad \beta = 1/2$

$$w_{i+1} = w_i + hf\left(w_i + \frac{h}{2}f(w_i, t_i), t_i + \frac{h}{2}\right)$$



Global error  
 $O(h^2)$

Better than Euler's

# Runge - Kutta Methods

Methods that use intermediate slopes to improve the global error.

⚡  
↓

To get  $w_{i+1}$  (at  $t_{i+1}$ )  
evaluate  $f(y, t)$  at intermediate  $y$   
and  $t \in [t_i, t_{i+1}]$ .

Examples:

- Ralston's method  $\rightsquigarrow O(h^2)$
- Heun's method
- Kutta's 3<sup>rd</sup> order
- Classical Runge-Kutta  $\rightsquigarrow O(h^4)$

}  $O(h^3)$

# Runge-Kutta Methods

the order is different  
than my notation!!

**Ralston's 2nd-order method** Local error  $O(h^3)$ .

$$w_{i+1} = w_i + \frac{1}{4}h_i \left( f(t_i, w_i) + 3f\left(t_i + \frac{2}{3}h_i, w_i + \frac{2}{3}h_i f(t_i, w_i)\right) \right).$$

$$k_{i,1} = h_i f(t_i, w_i); k_{i,2} = h_i f\left(t_i + \frac{2}{3}h_i, w_i + \frac{2}{3}k_{i,1}\right); w_{i+1} = w_i + \frac{1}{4}(k_{i,1} + 3k_{i,2}).$$

**Heun's 3rd-order method** Local error  $O(h^4)$ .

$$k_{i,1} = h_i f(t_i, w_i); k_{i,2} = h_i f\left(t_i + \frac{1}{3}h_i, w_i + \frac{1}{3}k_{i,1}\right); k_{i,3} = h_i f\left(t_i + \frac{2}{3}h_i, w_i + \frac{2}{3}k_{i,2}\right);$$

$$w_{i+1} = w_i + \frac{1}{4}(k_{i,1} + 3k_{i,3}).$$

**Kutta's 3rd-order method** Local error  $O(h^4)$ .

$$k_{i,1} = h_i f(t_i, w_i); k_{i,2} = h_i f\left(t_i + \frac{1}{2}h_i, w_i + \frac{1}{2}k_{i,1}\right); k_{i,3} = h_i f(t_i + h_i, w_i - k_{i,1} + 2k_{i,2});$$

$$w_{i+1} = w_i + \frac{1}{6}(k_{i,1} + 4k_{i,2} + k_{i,3}).$$

**Classical 4th-order Runge-Kutta method** Local error  $O(h^5)$ .

224  $k_{i,1} = h_i f(t_i, w_i); k_{i,2} = h_i f\left(t_i + \frac{1}{2}h_i, w_i + \frac{1}{2}k_{i,1}\right);$

$$k_{i,3} = h_i f\left(t_i + \frac{1}{2}h_i, w_i + \frac{1}{2}k_{i,2}\right); k_{i,4} = h_i f(t_i + h_i, w_i + k_{i,3});$$

$$w_{i+1} = w_i + \frac{1}{6}(k_{i,1} + 2k_{i,2} + 2k_{i,3} + k_{i,4}).$$

→ global error  
 $O(h^4)$

Kutta's method is generally better because the linear combination is symmetrical.

Example:  $\dot{y} = f(y, t) = e^{-t} - y^2$  w/  $\underline{y(0)=0}$

we want  $y(1)$ . use  $h=0.5$   $y(t_0)=y_0$   
 $\downarrow$   $\downarrow$   
 $0$   $0$

Use RK4

$$w_{i+1} = w_i + \frac{1}{6} (k_{i,1} + 2k_{i,2} + 2k_{i,3} + k_{i,4})$$

$$t_0 = 0 \quad w_0 = y(t_0) = y(0) = 0$$

$$k_{0,1} = h f(w_0, t_0) = 0.5 (e^0 - 0^2) = 0.5$$

$$k_{0,2} = h f(t_0 + \frac{1}{2}h, \frac{1}{2}k_{0,1}) = 0.5 f\left(0 + \frac{1}{2} \cdot 0.5, \frac{1}{2} \cdot 0.5\right)$$

$$= 0.5 \times 0.7163 = 0.3582$$

$$k_{0,3} = \dots = 0.1456 \quad k_{0,4} = \dots = 0.0530$$

$$k_0 = \frac{1}{6} (k_{0,1} + 2k_{0,2} + 2k_{0,3} + k_{0,4}) = \dots = 0.3661$$

$$w_1 = w_0 + k_0 = 0.3661$$

$\vdots$

$$w_2 = 0.5025$$

$$\text{absolute error} \approx 0.00085$$

much better than Euler! AS expected!

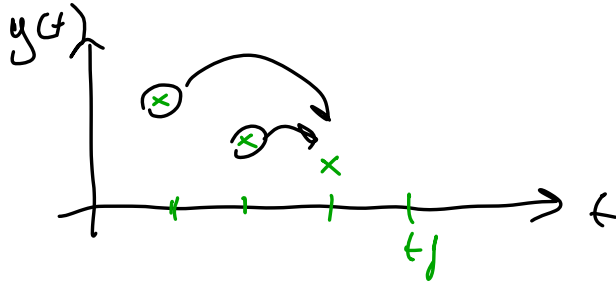
## RK methods

Until order 4 you need  $p$  function evaluations  
for a  $p^{\text{th}}$  order method

Higher  $p$   $\leadsto$  many more than  $p$  number of evaluations

4<sup>th</sup> order is the sweet-spot . . .

## multistep methods



use  $w_{i-n+1}, \dots, w_i$   
to determine  $w_{i+1}$

---

n-step method

$$w_{i+1} = \sum_{j=0}^{n-1} \alpha_j \underline{w_{i-j}} + h \sum_{j=0}^{n-1} \underline{b_j} \underline{f(w_{i-j}, t_{i-j})}$$

I know!  
from previous steps

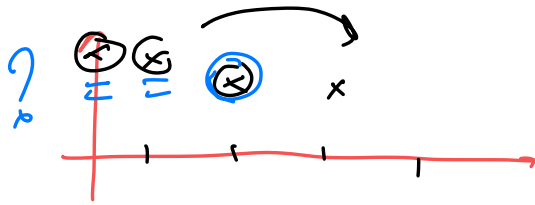
# Multistep methods

**2-stage Adams-Bashforth method** Local error  $O(h^3)$ .

$$w_{i+1} = w_i + (h/2)(3f(t_i, w_i) - f(t_{i-1}, w_{i-1})).$$

**3-stage Adams-Bashforth method** Local error  $O(h^4)$ .  $\rightarrow$  g.e.  $O(h^3)$

$$w_{i+1} = w_i + (h/12)(23f(t_i, w_i) - 16f(t_{i-1}, w_{i-1}) + 5f(t_{i-2}, w_{i-2})).$$



**Bootstrapping:** Use a one-step method to get needed values.

~~Match~~ match the order of  
one-step method w/  
multi-step method.

use a  
same or  
a higher-order  
method